Introduction

Tree

Data Structures and its Applications (UE25CS52A)

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Quiz : Heap

1. Searching for a key in a maxheap in the worst case makes

a) n comparisons

b) n – 1 comparisons

c) floor(n/2) comparisons

d) ceil(n/2) comparisons

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Quiz : Heap

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Quiz : Heap

2. adding a key in a maxheap in the worst case makes

a) n operations

b) n – 1 operations

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d) ceil(n/2) operations

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Quiz : Heap

2. adding a key in a maxheap in the worst case makes

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Quiz : Heap

3. Finding the maximum key in a maxheap in the worst case makes

a) n operations

b) n – 1 operations

c) 1 operation

d) n/2 operations

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Quiz : Heap

3. Finding the maximum key in a maxheap in the worst case makes

a) n operations

b) n – 1 operations

c) 1 operation

d) n/2 operations

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Quiz : Heap

4. What happens if the left subtree and the right subtree at each node are swapped?

a) The tree is no more a heap

b) The max heap becomes a heap tree

c) The max heap remains to be a max heap

d) Depends on the keys in the tree

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Quiz : Heap

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Quiz : Heap

5. A heap has n elements with its root at index 1. Which condition will tell us that the ith node does not have a right subtree?

a) 2i + 1 > n

b) 2i > n

c) 2i + 1 >= n

d) 2i >= n

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Quiz : Heap

5. A heap has n elements with its root at index 1. Which condition will tell us that the ith node does not have a right subtree?

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Heap

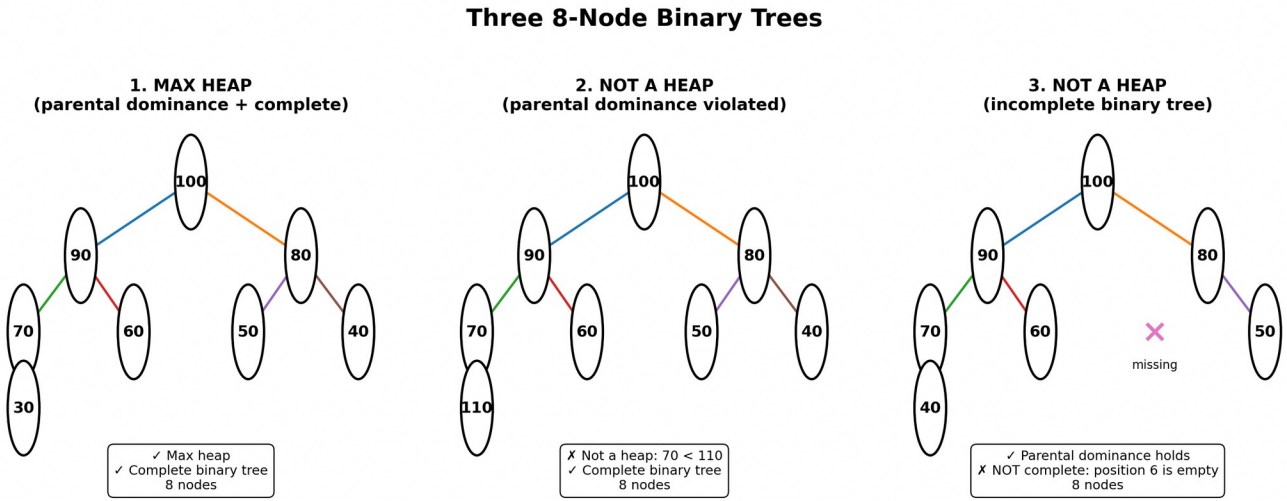
A heap(maxheap) is a tree with two special properties.

1. Complete binary tree:
   The nodes are filled up at each level.
   At the bottom most level, leftmost nodes are filled up.

2. Parental dominance:
   The key at the root is greater than the keys of its subtrees recursively.

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Heap



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Heap

Heap Implementation:

- Always as an array
- Method 1:
    - Start from index 0
    - Leftchild(i) = 2i + 1
    - Rightchild(i) = 2i + 2
    - Parent(i) = floor((i-1)/2)
- Method 2:
    - Start from index 1
    - Leftchild(i) = 2i
    - Rightchild(i) = 2i + 1
    - Parent(i) = floor(i/2)
- We will use the method 2

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Heap

Advantages of heap:

- No pointers
- No extra storage
- Easy way to determine empty or full
- Simple way to traverse either as an array or as a tree
- Does not waste memory as the tree is a complete binary tree

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Heap

uses of heap:

- Sorting : not commonly used
- Finding max(or min) repeatedly
  - Example: priority queue
  - Example: n way merge

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Heap

Conversion of a given array to a heap

- Refer heap1 folder; heap.h, heap.c and client.c
- Heap structure:

```
struct heap
{
        int key_[MAXSIZE];
        int n_;
};
typedef struct heap  heap_t;
```


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Heap

Conversion of a given array to a heap

- Copy the given array to the heap
- Leaves are always heaps
- From the first internal node, till the root, move indices backwards
    - Make each subtree a heap(heapify)

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Heap

Conversion of a given array to a heap

- Heapify:
- Make the heap of n nodes heapify
  about the ith node

```
static void heapify(int a[], int n, int pos)
{
        int i = pos;
        int j = 2 * i;
        int val = a[i];
        int is_heap = 0;
        while(j <= n && ! is_heap){
                if(j + 1 <= n && a[j+1] > a[j])
                        ++j;
                if(a[j] > val) {
                        a[i] = a[j];
                        i = j;
                        j = 2 * i;
                }
                else {
                        is_heap = 1;
                }
        }
a[i] = val;
}
```


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Heap

Implement insert a node at the end and remove the maximum

- Refer the folder heap2 – heap.h heap.c
- void insert(heap_t *ptr_heap, int key);
- int delete_max(heap_t *ptr_heap);
-

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Heap

Implement insert a node at the end and remove the maximum

- Refer the folder heap2 – heap.h heap.c
- Insert : add at the end; check whether the parent tree is a heap repeatedly until the root is reached

```
void insert(heap_t *ptr_heap, int key)
{
        int pos = ++ptr_heap->n_;
                ptr_heap->key_[pos] = key;
        while(pos > 1 && ptr_heap->key_[pos] > ptr_heap->key_[pos/2])
        {
                swap(&ptr_heap->key_[pos], &ptr_heap->key_[pos/2]);
                pos = pos / 2;
        }
}
```


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Heap

Implement insert a node at the end and remove the maximum

- Refer the folder heap2 – heap.h heap.c
- Delete max : replace the first node with the last and heapify about the root.

```
int delete_max(heap_t *ptr_heap)
{
        int res = ptr_heap->key_[1];
        ptr_heap->key_[1] = ptr_heap->key_[ptr_heap->n_];
        heapify(ptr_heap->key_, --ptr_heap->n_, 1);
        return res;
}
```


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Heap

Implement a priority queue using heap

- Refer the folder priority queue –queue.h queue.c heap.h heap.c
- The queue data structure embeds the heap. The functions of the priority queue like enqueue and dequeue are forwarded to the embedded heap.

```
struct node
{
        int key_;
        struct node* next_;
};
typedef struct node node_t;
 struct queue
{
        heap_t q_;
};
typedef struct queue queue_t;
```


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Heap

Implement a priority queue using heap

```
void enqueue(queue_t *ptr_queue, int key)
{
        insert(&ptr_queue->q_, key);

}
int dequeue(queue_t *ptr_queue)
{
        return delete_max(&ptr_queue->q_);
}
```


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Heap

Implement a priority queue using heap

- In other implementations of priority queue, one of the operations(enqueue or dequeue) would not be that efficient – the number of operations would depend on the number of elements in the queue
- In this implementation, both the operations are proportional to log(n, 2) where is the number of elements

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THANK YOU

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